

计算机图形学作业报告 week

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算法描述与核心代码讲解

实验结果

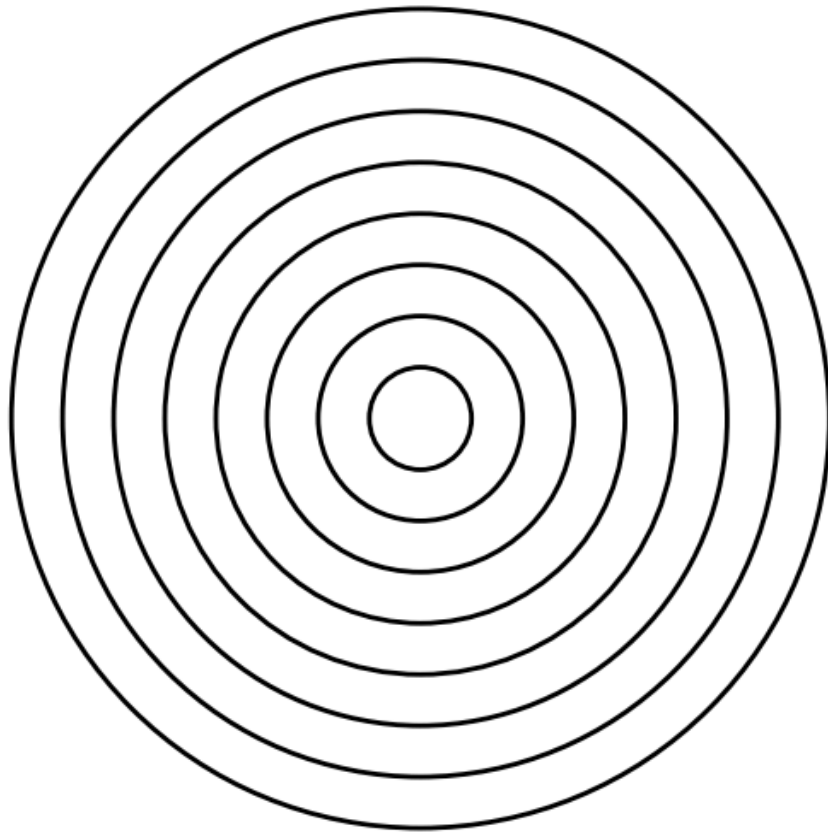
配置环境:

```
pip install -r requirements.txt
```

运行程序:

```
./run.bat
```

实验结果如下: (可插入图片)



参考

1. <https://chatgpt.com/>

源文件

在这里粘贴上所有源代码

concentric_circles.py

```
"""使用 Matplotlib 绘制同心圆。"""

from pathlib import Path
import sys

import matplotlib

matplotlib.use("Agg")
```

```

import matplotlib.pyplot as plt
from matplotlib.patches import Circle

FIGURE_SIZE = 7
FIGURE_DPI = 100
WEEK2_IMAGE_DIRECTORY = Path(__file__).resolve().parent.parent / "docs" /
"images"

from image_utils import save_figure

def draw_concentric_circles(
    count: int = 8,
    spacing: float = 30,
) -> Path:
    """绘制并保存一组圆心相同、半径等间距的圆。"""
    if count <= 0:
        raise ValueError("count 必须大于 0")
    if spacing <= 0:
        raise ValueError("spacing 必须大于 0")

    figure, axes = plt.subplots(
        figsize=(FIGURE_SIZE, FIGURE_SIZE),
        dpi=FIGURE_DPI,
    )
    figure.canvas.manager.set_window_title("Concentric Circles")

    for index in range(1, count + 1):
        radius = index * spacing
        axes.add_patch(
            Circle(
                (0, 0),
                radius,
                fill=False,
                linewidth=2,
            )
        )

    outer_radius = count * spacing
    limit = outer_radius + spacing
    axes.set_xlim(-limit, limit)
    axes.set_ylim(-limit, limit)
    axes.set_aspect("equal", adjustable="box")
    axes.axis("off")

    saved_path = save_figure(
        figure,
        "concentric_circles.png",
        output_directory=WEEK2_IMAGE_DIRECTORY,
    )
    print(f"Saved image: {saved_path}")

```

```
plt.close(figure)

return saved_path

if __name__ == "__main__":
    draw_concentric_circles()
```

image_utils.py复用了week1的文件。